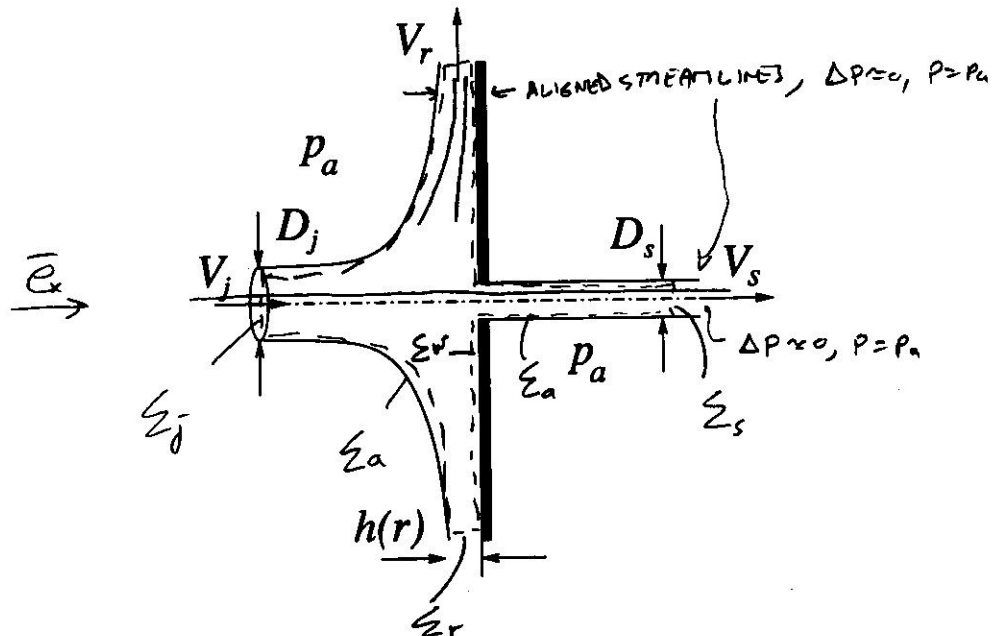


CL8

A circular water jet of diameter D_j and velocity V_j impinges perpendicularly to a wall, where there exists an orifice centered at the jet axis, as shown in the figure, so that part of the water flows through the orifice and part is deflected to flow next to the wall. Assuming that the flow velocity is sufficiently large for the effects of viscosity and gravity to be negligible:

1. Determine the values of the velocities V_s and V_r for the secondary jet that emerges from the orifice and for the radial wall jet.
2. Obtain the thickness $h(r)$ of the radial wall jet.
3. Compute the force that the jet exerts on the wall as a function of ρ , V_j , D_j and D_s .



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$$P + \frac{1}{2} \rho V^2 = C_p \Rightarrow \begin{aligned} P_a + \frac{1}{2} \rho V_j^2 &= P_a + \frac{1}{2} \rho V_r^2 \rightarrow V_j = V_r \\ P_a + \frac{1}{2} \rho V_j^2 &= P_a + \frac{1}{2} \rho V_s^2 \rightarrow V_j = V_s \end{aligned}$$

CONTINUITY $-\rho V_j \frac{\pi D_j^2}{4} + \rho V_s \frac{\pi D_s^2}{4} + 2\pi r h V_r = 0 \Rightarrow h = \frac{D_j^2 - D_s^2}{8r}$

$\vec{F}_w = - \int_{\Sigma_2} P_a \vec{n} d\sigma - \int_{\Sigma_1} P \vec{n} d\sigma + \int_{\Sigma_1} \vec{\tau} \cdot \vec{n} d\sigma = - \int_{\Sigma_1} (P - P_a) \vec{n} d\sigma$

MOMENTUM APPLIED TO THE CONTROL VOLUME

$$\frac{d}{dt} \int_{V_c} \rho \vec{v} dV + \int_{\Sigma_2 + \Sigma_r + \Sigma_s + \Sigma_a + \Sigma_j} \rho \vec{v} (\vec{v} \cdot \vec{n}) \cdot \vec{n} d\sigma = - \int_{\Sigma_1} (P - P_a) \vec{n} d\sigma$$

$$\left(\rho V_s^2 \frac{\pi D_s^2}{4} - \rho V_j^2 \frac{\pi D_j^2}{4} \right) \vec{e}_x = \vec{F}_w \rightarrow \vec{F}_w = \rho V_j^2 \frac{\pi}{4} (D_j^2 - D_s^2)$$